

Zero Trust SOC Lab Incident Report

Incident Summary

This report documents a simulated reconnaissance attack within a Zero Trust network environment. The activity was detected and blocked using pfSense firewall rules and monitored via Splunk SIEM.

Environment

- 1 Attacker: Kali Linux (192.168.2.100)
- 2 Firewall: pfSense
- 3 SIEM: Splunk Enterprise
- 4 Target Network: LAN (192.168.1.0/24)
- 5 DMZ Network: 192.168.2.0/24

Incident Classification

- 1 Type: Reconnaissance / Network Scanning
- 2 Severity: Medium
- 3 MITRE ATT&CK: T1595 – Active Scanning

Attack Details

Technique: Nmap SYN Scan

Command Used:

```
nmap -Pn -sS 192.168.1.0/24
```

Detection

The activity was detected in Splunk using the following query, which identified repeated blocked connection attempts from a single source IP, indicating scanning behavior:

```
index=main "UFW BLOCK"  
| stats count by SRC
```

Evidence

- 1 Multiple blocked attempts from source IP: 192.168.2.100
- 2 Targeted multiple internal hosts and ports
- 3 Confirmed via pfSense firewall logs and Splunk SIEM logs
- 4 Visualized using Splunk dashboard

Timeline

- 1 Attack Start: 2026-04-28 14:23
- 2 Detection: Real-time via Splunk
- 3 Response: Immediate (firewall block)

Response

- 1 Zero Trust firewall rules blocked DMZ to LAN traffic
- 2 Logs ingested into Splunk for monitoring and analysis
- 3 Dashboard used to visualize attack patterns

Impact

No successful intrusion occurred. No lateral movement was observed. All malicious traffic was blocked.

Conclusion

The Zero Trust architecture successfully prevented unauthorized access and lateral movement. The SOC workflow effectively detected, analyzed, and responded to the attack.